

ECON 320 — LECTURE NOTES

Sets and Functions

Mathematical Foundations for Economic Analysis

Course: ECON 320: Introduction to Mathematical Economics

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1 Introduction to Sets

1.1 What is a Set?

A **set** is one of the most primitive objects in mathematics. At its core, a set is simply a well-defined collection of distinct objects. “Well-defined” means that for any object, we can determine with certainty whether or not it belongs to the collection.

Definition: Set

A **set** A is a collection of distinct objects called **elements** (or **members**). We write $x \in A$ to mean “ x is an element of A ”, and $x \notin A$ to mean “ x is not an element of A ”.

Sets are typically denoted by capital letters $A, B, C, \dots$ and their elements by lower-case letters $a, b, c, \dots$. There are two standard ways to describe a set:

1. **Roster (Enumeration) notation:** list all elements between braces.
 $A = \{1, 2, 3, 4\}, \quad B = \{a, e, i, o, u\}$
2. **Set-builder (Predicate) notation:** specify a rule that elements must satisfy.
 $C = \{x \in \mathbb{R} : x^2 < 4\} = (-2, 2), \quad D = \{x \in \mathbb{N} : x \text{ is even}\}$

Definition: Special Sets

The following sets appear constantly in economic analysis:

- $\mathbb{N} = \{1, 2, 3, \dots\}$ Natural numbers
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ Integers
- $\mathbb{Q} = \left\{\frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0\right\}$ Rational numbers
- $\mathbb{R}$ Real numbers (the real line)
- $\mathbb{R}^n = \{(x_1, \dots, x_n) : x_i \in \mathbb{R}\}$ n -dimensional Euclidean space
- $\emptyset = \{\}$ The empty set (contains no elements)

Worked Example: Economic Sets

In economics, sets arise naturally:

- **Commodity space:** $\mathbb{R}_+^n = \{(x_1, \dots, x_n) \in \mathbb{R}^n : x_i \geq 0 \forall i\}$ — the set of all non-negative consumption bundles of n goods.
- **Budget set:** $B(p, w) = \{x \in \mathbb{R}_+^n : p \cdot x \leq w\}$ — bundles affordable at prices p and income w .
- **Production set:** $Y \subseteq \mathbb{R}^n$ — the set of all technically feasible input-output vectors.
- **Feasible set** in a maximisation problem: $\{x \in \mathbb{R}^n : g(x) \leq b\}$.

1.2 Set Membership and Equality

Definition: Set Equality

Two sets A and B are **equal**, written $A = B$, if and only if they contain exactly the same elements:

$$A = B \iff (x \in A \iff x \in B)$$

Three important observations about sets:

- **Order does not matter:** $\{1, 2, 3\} = \{3, 1, 2\}$.
- **Repetition does not matter:** $\{1, 1, 2, 3\} = \{1, 2, 3\}$.
- **The empty set is unique:** there is only one empty set.

Worked Example: Verifying Set Equality

Are $A = \{x \in \mathbb{R} : x^2 - 5x + 6 = 0\}$ and $B = \{2, 3\}$ equal?
 Solve $x^2 - 5x + 6 = 0$: $(x - 2)(x - 3) = 0 \Rightarrow x = 2$ or $x = 3$.
 So $A = \{2, 3\} = B$. ✓

1.3 Subsets and the Power Set

Definition: Subset

A is a **subset** of B , written $A \subseteq B$, if every element of A is also an element of B :

$$A \subseteq B \iff (x \in A \Rightarrow x \in B)$$

A is a **proper subset** of B , written $A \subsetneq B$, if $A \subseteq B$ and $A \neq B$.

Common Error: $\in$ versus $\subseteq$

These symbols are often confused. $\in$ relates an *element* to a set; $\subseteq$ relates a *set* to a set.

- Correct: $2 \in \{1, 2, 3\}$ and $\{2\} \subseteq \{1, 2, 3\}$
- Wrong: $2 \subseteq \{1, 2, 3\}$ or $\{2\} \in \{1, 2, 3\}$
- Note: $\{2\}$ is *not* an element of $\{1, 2, 3\}$ — it would need to be listed as an element.

2 Set Operations

2.1 The Four Basic Operations

Definition: Union

The **union** of A and B is the set of elements belonging to A or B (or both):

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

Definition: Intersection

The **intersection** of A and B is the set of elements belonging to both A and B :

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

If $A \cap B = \emptyset$, the sets are called **disjoint**.

Definition: Set Difference and Complement

The **set difference** (or relative complement) of B from A :

$$A \setminus B = A - B = \{x : x \in A \text{ and } x \notin B\}$$

If all sets under discussion are subsets of a **universal set** U , the **complement** of A is:

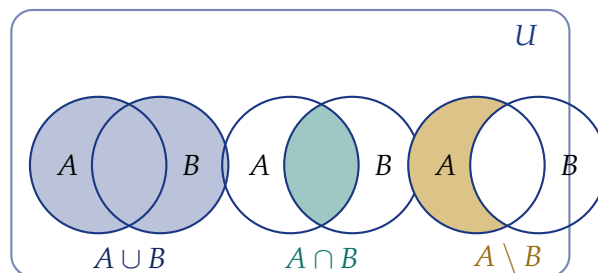
$$A^c = U \setminus A = \{x \in U : x \notin A\}$$

Definition: Cartesian Product

The **Cartesian product** of A and B is the set of all ordered pairs:

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

More generally, $A_1 \times \cdots \times A_n = \{(a_1, \dots, a_n) : a_i \in A_i\}$. Note $\mathbb{R}^n = \underbrace{\mathbb{R} \times \cdots \times \mathbb{R}}_n$.



Venn diagrams: Union (blue), Intersection (teal), Set Difference (gold)

Worked Example: Set Operations in Consumer Theory

Let the universal set be $U = \mathbb{R}_+^2$ (all non-negative bundles of two goods). Let:

$$A = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1 + x_2 \leq 10\} \quad (\text{affordable at price 1, income 10})$$

$$B = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1 \geq 2 \text{ and } x_2 \geq 2\} \quad (\text{minimum requirements})$$

Then:

- $A \cup B$ = bundles that are either affordable *or* meet minimum requirements.
- $A \cap B = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1 + x_2 \leq 10, x_1 \geq 2, x_2 \geq 2\}$ = affordable bundles that also meet minimum requirements (the relevant feasible set).
- $A \setminus B = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1 + x_2 \leq 10\} \setminus B$ = affordable bundles that *fail* to meet at least one minimum requirement.
- $A^c = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1 + x_2 > 10\}$ = unaffordable bundles.

2.2 Laws of Set Algebra

The following identities hold for all sets A, B, C within universal set U :

Theorem: Laws of Set Algebra

$$\text{Commutativity: } A \cup B = B \cup A \qquad A \cap B = B \cap A$$

$$\text{Associativity: } (A \cup B) \cup C = A \cup (B \cup C) \qquad (A \cap B) \cap C = A \cap (B \cap C)$$

$$\text{Distributivity: } A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \qquad A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$\text{Identity: } A \cup \emptyset = A \qquad A \cap U = A$$

$$\text{Complement: } A \cup A^c = U \qquad A \cap A^c = \emptyset$$

$$\text{De Morgan's Laws: } (A \cup B)^c = A^c \cap B^c \qquad (A \cap B)^c = A^c \cup B^c$$

$$\text{Double complement: } (A^c)^c = A$$

Worked Example: Proving a Set Identity

Prove: $A \setminus B = A \cap B^c$.

Proof: We show the two sets have identical membership.

($\subseteq$) Let $x \in A \setminus B$. Then $x \in A$ and $x \notin B$. Since $x \notin B$, we have $x \in B^c$. Therefore $x \in A \cap B^c$.

($\supseteq$) Let $x \in A \cap B^c$. Then $x \in A$ and $x \in B^c$, i.e. $x \in A$ and $x \notin B$. Therefore $x \in A \setminus B$.

Since each set is a subset of the other, $A \setminus B = A \cap B^c$. □

Worked Example: Applying De Morgan's Laws in Economics

The **infeasible** region is the complement of the feasible set. Suppose the feasible set is:

$$F = \{x \in \mathbb{R}^n : g_1(x) \leq b_1\} \cap \{x \in \mathbb{R}^n : g_2(x) \leq b_2\} = G_1 \cap G_2$$

By De Morgan's Law:

$$F^c = (G_1 \cap G_2)^c = G_1^c \cup G_2^c$$

A bundle is *infeasible* if and only if it violates *at least one* constraint — which is exactly $G_1^c \cup G_2^c$ (the union of the individual violation sets).

2.3 Indexed Families of Sets

When working with many sets simultaneously, we use indexed notation.

Definition: Indexed Union and Intersection

Let $\{A_\alpha\}_{\alpha \in I}$ be a family of sets indexed by I . Then:

$$\bigcup_{\alpha \in I} A_\alpha = \{x : x \in A_\alpha \text{ for some } \alpha \in I\}$$

$$\bigcap_{\alpha \in I} A_\alpha = \{x : x \in A_\alpha \text{ for all } \alpha \in I\}$$

Worked Example: Nested Budget Sets

For each income level $w > 0$, define the budget set $B_w = \{x \in \mathbb{R}_+^2 : p_1x_1 + p_2x_2 \leq w\}$ at fixed prices (p_1, p_2) .

- $\bigcup_{w>0} B_w = \mathbb{R}_+^2$ (any bundle is affordable at high enough income)
- $\bigcap_{w>0} B_w = \{(0,0)\}$ (only the origin is affordable at every income level)
- If $w_1 < w_2$, then $B_{w_1} \subsetneq B_{w_2}$ (nested structure: richer budget sets contain poorer ones)

3 Real Analysis: Intervals and Bounded Sets**3.1 Intervals in $\mathbb{R}$**

Intervals are the most important subsets of $\mathbb{R}$ in economic analysis. They model price ranges, income levels, time horizons, and parameter spaces.

Definition: Types of Intervals

For $a, b \in \mathbb{R}$ with $a < b$:

$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$	closed interval
$(a, b) = \{x \in \mathbb{R} : a < x < b\}$	open interval
$[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$	half-open (closed at a)
$(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$	half-open (closed at b)
$[a, \infty) = \{x \in \mathbb{R} : x \geq a\}$	closed ray
$(a, \infty) = \{x \in \mathbb{R} : x > a\}$	open ray
$\mathbb{R} = (-\infty, \infty)$	entire real line

Definition: Bounded Sets

A set $A \subseteq \mathbb{R}$ is:

- **Bounded above** if there exists $M \in \mathbb{R}$ such that $x \leq M$ for all $x \in A$. The smallest such M is the **supremum** $\sup A$.
- **Bounded below** if there exists $m \in \mathbb{R}$ such that $x \geq m$ for all $x \in A$. The largest such m is the **infimum** $\inf A$.
- **Bounded** if it is bounded both above and below.

If $\sup A \in A$, it is called the **maximum** $\max A$. If $\inf A \in A$, it is called the **minimum** $\min A$.

Worked Example: Supremum vs. Maximum

1. $A = [0, 1]$: $\sup A = 1 = \max A$ and $\inf A = 0 = \min A$. (Both attained.)
2. $B = (0, 1)$: $\sup B = 1$ but $\max B$ does not exist (1 is not in B). Similarly $\inf B = 0$, $\min B$ does not exist.
3. $C = \{1/n : n \in \mathbb{N}\} = \{1, 1/2, 1/3, \dots\}$: $\sup C = 1 = \max C$, but $\inf C = 0$ and $\min C$ does not exist (0 is not in C).

Economic relevance: In maximisation problems, we need $\sup$ when the maximum may not be attained (e.g. open feasible sets). The Extreme Value Theorem guarantees $\max$ exists when the domain is *closed and bounded* (compact) and the objective is continuous.

3.2 Open and Closed Sets in $\mathbb{R}^n$ **Definition: Open Ball, Open Set, Closed Set**

In $\mathbb{R}^n$, the **open ball** of radius $r > 0$ centred at x_0 is:

$$B_r(x_0) = \{x \in \mathbb{R}^n : \|x - x_0\| < r\}$$

A set $U \subseteq \mathbb{R}^n$ is **open** if for every $x \in U$, there exists $r > 0$ such that $B_r(x) \subseteq U$.

A set $F \subseteq \mathbb{R}^n$ is **closed** if its complement $F^c = \mathbb{R}^n \setminus F$ is open.

Key Facts About Open and Closed Sets

- $\emptyset$ and $\mathbb{R}^n$ are both open *and* closed.
- Finite intersections and arbitrary unions of open sets are open.
- Finite unions and arbitrary intersections of closed sets are closed.
- A set can be neither open nor closed (e.g. $[0, 1)$ in $\mathbb{R}$).
- The budget set $\{x \geq 0 : p \cdot x \leq w\}$ is closed; the strict budget set $\{x \geq 0 : p \cdot x < w\}$ is open.

4 Functions: Core Concepts

4.1 Definition and Notation

Definition: Function

A **function** f from set X to set Y , written $f : X \rightarrow Y$, is a rule that assigns to each element $x \in X$ **exactly one** element $f(x) \in Y$.

- X is the **domain** of f : $\text{dom}(f) = X$
- Y is the **codomain** of f
- $f(x)$ is the **image** (or value) of x under f
- The **range** (image) of f : $\text{ran}(f) = \text{img}(f) = \{f(x) : x \in X\} \subseteq Y$

Functions are not just formulas

A function is fully specified by its domain, codomain, and rule. The same formula with different domains defines different functions:

- $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$ vs. $g : [0, \infty) \rightarrow \mathbb{R}, g(x) = x^2$

These are *different* functions: g is invertible while f is not. Always state the domain explicitly.

Worked Example: Domain and Range in Economics

Consider the inverse demand function $f : (0, \infty) \rightarrow (0, \infty), f(Q) = \frac{100}{Q}$.

- Domain: $\text{dom}(f) = (0, \infty)$ (quantity must be strictly positive).
- Range: $\text{ran}(f) = (0, \infty)$ (price takes all positive values).

Now consider a Cobb-Douglas production function $F : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+, F(K, L) = K^\alpha L^{1-\alpha}$ for $\alpha \in (0, 1)$.

- Domain: $\mathbb{R}_+^2$ (capital and labour must be non-negative).
- Range: $\mathbb{R}_+ = [0, \infty)$.

4.2 Injective, Surjective, and Bijective Functions

Definition: Injective (One-to-One)

$f : X \rightarrow Y$ is **injective** (one-to-one) if distinct inputs produce distinct outputs:

$$f(x_1) = f(x_2) \implies x_1 = x_2 \quad (\text{equivalently: } x_1 \neq x_2 \implies f(x_1) \neq f(x_2))$$

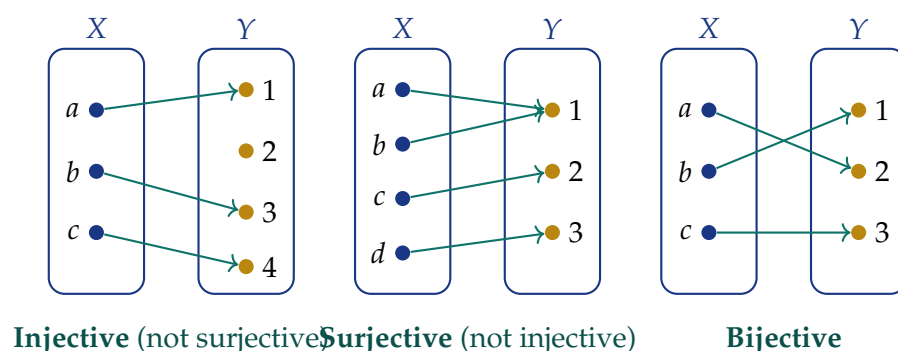
Definition: Surjective (Onto)

$f : X \rightarrow Y$ is **surjective** (onto) if every element of the codomain is hit by at least one input:

$$\forall y \in Y, \exists x \in X \text{ s.t. } f(x) = y \quad (\text{equivalently: } \text{ran}(f) = Y)$$

Definition: Bijective

$f : X \rightarrow Y$ is **bijective** (a one-to-one correspondence) if it is both injective and surjective. Bijections set up a perfect pairing between X and Y .



Worked Example: Classifying Economic Functions

1. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 3$. **Bijective.**
 - Injective: $f(x_1) = f(x_2) \Rightarrow 2x_1 + 3 = 2x_2 + 3 \Rightarrow x_1 = x_2$. ✓
 - Surjective: for any y , $x = (y - 3)/2 \in \mathbb{R}$ satisfies $f(x) = y$. ✓
2. $g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = x^2$. **Neither injective nor surjective.**
 - Not injective: $g(-2) = g(2) = 4$ but $-2 \neq 2$.
 - Not surjective: $-1 \in \mathbb{R}$ but $g(x) = x^2 \geq 0 > -1$ for all x , so no x maps to -1 .
3. $h : [0, \infty) \rightarrow [0, \infty), h(x) = x^2$. **Bijective.**
 - Injective: $h(x_1) = h(x_2)$ and $x_1, x_2 \geq 0 \Rightarrow x_1 = x_2$. ✓
 - Surjective: for any $y \geq 0$, $x = \sqrt{y} \geq 0$ satisfies $h(x) = y$. ✓
4. Profit function $\pi : \mathbb{R}_+ \rightarrow \mathbb{R}, \pi(q) = pq - cq^2$. **Not injective** (two output levels can yield the same profit), **not surjective onto** $\mathbb{R}$ from $\mathbb{R}_+$ (range is $(-\infty, p^2/(4c)]$).

4.3 Image and Pre-image

Definition: Image and Pre-image of Sets

Let $f : X \rightarrow Y$, $A \subseteq X$, and $B \subseteq Y$.

The **image** of A under f :

$$f(A) = \{f(x) : x \in A\} \subseteq Y$$

The **pre-image** (inverse image) of B under f :

$$f^{-1}(B) = \{x \in X : f(x) \in B\} \subseteq X$$

Note: $f^{-1}(B)$ is defined for any function f (even non-invertible ones); it simply collects all inputs mapping into B .

Theorem: Properties of Images and Pre-images

Let $f : X \rightarrow Y$, $A_1, A_2 \subseteq X$, and $B_1, B_2 \subseteq Y$. Then:

$$f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$$

$$f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2) \quad (\text{equality holds if } f \text{ is injective})$$

$$f^{-1}(B_1 \cup B_2) = f^{-1}(B_1) \cup f^{-1}(B_2)$$

$$f^{-1}(B_1 \cap B_2) = f^{-1}(B_1) \cap f^{-1}(B_2)$$

$$f^{-1}(B^c) = (f^{-1}(B))^c$$

Pre-images behave better than images under set operations.

Worked Example: Pre-image in Welfare Analysis

Let $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ be a utility function $f(x) = \ln(1 + x)$. The regulator defines a welfare threshold $B = [\ln 2, \infty)$ (utility at least $\ln 2$).

The pre-image is:

$$f^{-1}(B) = \{x \geq 0 : \ln(1 + x) \geq \ln 2\} = \{x \geq 0 : 1 + x \geq 2\} = [1, \infty)$$

Interpretation: consumers with consumption $x \geq 1$ achieve the welfare threshold. The pre-image gives the *set of inputs that are socially adequate*.

5 Function Composition and Inverse Functions

5.1 Composition of Functions

Definition: Composition

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. The **composition** $g \circ f : X \rightarrow Z$ is defined by:

$$(g \circ f)(x) = g(f(x)) \quad \forall x \in X$$

Order of Composition

$g \circ f$ means “first apply f , then apply g ”. This is read right-to-left. In general, $g \circ f \neq f \circ g$ (composition is **not** commutative).

Worked Example: Composition in Macroeconomics

Let Y be GDP, U be unemployment, and I be inflation:

$$f : \text{GDP} \rightarrow \text{Unemployment}, \quad f(Y) = \bar{u} - \beta(Y - \bar{Y}) \quad (\text{Okun's Law})$$

$$g : \text{Unemployment} \rightarrow \text{Inflation}, \quad g(U) = \pi^e - \gamma(U - \bar{u}) \quad (\text{Phillips Curve})$$

The composed function $g \circ f : \text{GDP} \rightarrow \text{Inflation}$:

$$(g \circ f)(Y) = g(f(Y)) = \pi^e - \gamma(\bar{u} - \beta(Y - \bar{Y}) - \bar{u}) = \pi^e + \gamma\beta(Y - \bar{Y})$$

This is the “aggregate supply curve”: higher output above potential raises inflation via the unemployment channel.

Theorem: Properties of Composition

Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$, $h : Z \rightarrow W$. Then:

1. **Associativity:** $h \circ (g \circ f) = (h \circ g) \circ f$
2. If f and g are injective, then $g \circ f$ is injective.
3. If f and g are surjective, then $g \circ f$ is surjective.
4. If f and g are bijective, then $g \circ f$ is bijective and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

5.2 Inverse Functions

Definition: Inverse Function

Let $f : X \rightarrow Y$ be a bijection. The **inverse function** $f^{-1} : Y \rightarrow X$ is the unique function satisfying:

$$f^{-1}(f(x)) = x \quad \forall x \in X \quad \text{and} \quad f(f^{-1}(y)) = y \quad \forall y \in Y$$

Equivalently, $f^{-1} \circ f = \text{id}_X$ and $f \circ f^{-1} = \text{id}_Y$, where id denotes the identity function.

When Does an Inverse Exist?

$f : X \rightarrow Y$ has an inverse $f^{-1} : Y \rightarrow X$ if and only if f is **bijective**. For real-valued functions:

- A **strictly monotone** function $f : I \rightarrow \mathbb{R}$ on an interval I is injective.
- The inverse of a strictly increasing function is strictly increasing.
- The inverse of a strictly decreasing function is strictly decreasing.
- Geometrically: the graph of f^{-1} is the reflection of the graph of f across the line $y = x$.

Worked Example: Inverse Functions in Economics

1. Demand and Inverse Demand. Let $Q = D(P) = 120 - 2P$ be the demand function ($P \geq 0, Q \geq 0$).

- Domain of D : $[0, 60]$ (prices giving non-negative quantity).
- Inverse: solve for P : $Q = 120 - 2P \Rightarrow P = 60 - Q/2$.
- So $D^{-1}(Q) = P(Q) = 60 - Q/2$ is the inverse demand.
- This is economically meaningful: $P(Q)$ gives the maximum price consumers pay for Q units.

2. Expenditure and Indirect Utility. The expenditure function $e(p, u)$ and indirect utility function $v(p, w)$ are inverses of each other in the utility argument: $e(p, v(p, w)) = w$ and $v(p, e(p, u)) = u$. This is Shephard's duality.

3. Production. If $F : [0, \infty) \rightarrow [0, \infty)$, $F(L) = \sqrt{L}$, then $F^{-1}(Q) = Q^2$ gives the labour required to produce Q units.

Worked Example: Finding Inverse Functions: Step-by-Step

Find the inverse of $f(x) = \frac{3x-1}{x+2}$ (where $x \neq -2$).

Step 1: Write $y = \frac{3x-1}{x+2}$ and solve for x .

Step 2: $y(x+2) = 3x-1 \Rightarrow xy+2y = 3x-1 \Rightarrow xy-3x = -2y-1$

Step 3: $x(y-3) = -2y-1 \Rightarrow x = \frac{-2y-1}{y-3} = \frac{2y+1}{3-y}$

Step 4: So $f^{-1}(y) = \frac{2y+1}{3-y}$, defined for $y \neq 3$.

Verify: $f(f^{-1}(y)) = f\left(\frac{2y+1}{3-y}\right) = \frac{3 \cdot \frac{2y+1}{3-y} - 1}{\frac{2y+1}{3-y} + 2} = \frac{\frac{6y+3-(3-y)}{3-y}}{\frac{2y+1+2(3-y)}{3-y}} = \frac{7y}{7} = y \checkmark$

6 Types of Functions in Economic Analysis

6.1 Monotone Functions

Definition: Monotone Functions

A function $f : X \subseteq \mathbb{R} \rightarrow \mathbb{R}$ is:

- **Strictly increasing:** $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$
- **Weakly increasing:** $x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)$
- **Strictly decreasing:** $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$
- **Weakly decreasing:** $x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2)$

f is **monotone** if it is either weakly increasing or weakly decreasing. Strictly monotone functions are always injective.

6.2 Convex and Concave Functions

Definition: Convex and Concave Functions

A function $f : C \rightarrow \mathbb{R}$ on a convex set $C \subseteq \mathbb{R}^n$ is:

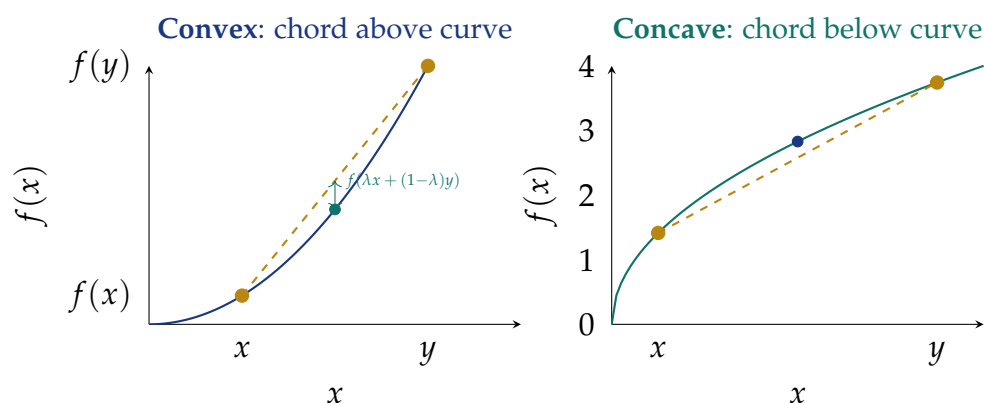
Convex if for all $x, y \in C$ and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

Concave if for all $x, y \in C$ and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$$

f is **strictly convex** (resp. **strictly concave**) if the inequality is strict for $\lambda \in (0, 1)$ and $x \neq y$.



Concavity in Economics

- **Utility functions** are typically assumed concave — this captures diminishing marginal utility and gives rise to risk aversion.
- **Production functions** are typically concave in inputs — this captures diminishing marginal product.
- **Cost functions** are convex in output — this reflects increasing marginal

cost.

- If f is concave and differentiable, then $f''(x) \leq 0$ for all x .
- Any local maximum of a concave function is a global maximum — critical for optimisation.

Worked Example: Testing Convexity and Concavity

1. $f(x) = x^2$, $f''(x) = 2 > 0$: **strictly convex**.
2. $f(x) = \ln x$ for $x > 0$, $f''(x) = -1/x^2 < 0$: **strictly concave**.
3. $f(x) = e^x$, $f''(x) = e^x > 0$: **strictly convex**.
4. Cobb-Douglas $F(K, L) = K^\alpha L^\beta$ with $\alpha + \beta < 1$: **concave** (diminishing returns to scale).

Verify using the definition for (2): For $x, y > 0$ and $\lambda \in [0, 1]$, we need $\ln(\lambda x + (1 - \lambda)y) \geq \lambda \ln x + (1 - \lambda) \ln y = \ln(x^\lambda y^{1-\lambda})$.

This says $\lambda x + (1 - \lambda)y \geq x^\lambda y^{1-\lambda}$, which is the **AM-GM inequality** — a classical result. ✓

6.3 Homogeneous Functions

Definition: Homogeneous Function

A function $f : \mathbb{R}_+^n \rightarrow \mathbb{R}$ is **homogeneous of degree k** if for all $t > 0$ and all $x \in \mathbb{R}_+^n$:

$$f(tx) = t^k f(x)$$

Theorem: Euler's Theorem

If $f : \mathbb{R}_+^n \rightarrow \mathbb{R}$ is differentiable and homogeneous of degree k , then:

$$\sum_{i=1}^n x_i \frac{\partial f}{\partial x_i}(x) = k \cdot f(x)$$

Worked Example: Homogeneity in Producer Theory

- **Constant returns to scale:** $F(tK, tL) = tF(K, L)$ — homogeneous of degree 1.
Euler: $K \cdot F_K + L \cdot F_L = F(K, L)$ — the sum of marginal products weighted by inputs equals total output (the zero-profit condition for competitive firms).
- **Cobb-Douglas** $F(K, L) = K^\alpha L^\beta$: $F(tK, tL) = (tK)^\alpha (tL)^\beta = t^{\alpha+\beta} K^\alpha L^\beta = t^{\alpha+\beta} F(K, L)$.
Degree = $\alpha + \beta$: IRS if > 1 , CRS if $= 1$, DRS if < 1 .
- **Demand functions:** Walrasian demand $x(p, w)$ is homogeneous of degree 0 in (p, w) — doubling all prices and income leaves demand unchanged (no money illusion).

7 Practice Questions

7.1 Section A: Sets and Set Operations

A1. (Set Builder Notation) Write each of the following sets in set-builder notation and determine whether it is finite or infinite:

- (a) The set of all even natural numbers less than 20.
- (b) The set of all real numbers whose square is less than 16.
- (c) The set of all prices in $[0, 100]$ that make demand positive, given $D(P) = 50 - P$.

A2. (Set Operations) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 5, 6, 7\}$, and $C = \{5, 6, 7, 8, 9\}$. Compute:

- (a) $A \cup B$ (b) $A \cap B$ (c) $A \setminus B$ (d) $B \setminus A$ (e) $(A \cup B)^c$ (f) $A \cap B \cap C$
(g) $A \cup (B \cap C)$

A3. (Power Set) Find the power set of $A = \{x, y\}$. How many elements does $\mathcal{P}(\mathcal{P}(A))$ have?

A4. (Cartesian Product) Let $A = \{1, 2\}$ and $B = \{a, b, c\}$. List all elements of $A \times B$ and $B \times A$. Are they equal?

A5. (De Morgan's Laws) Verify De Morgan's Law $(A \cup B)^c = A^c \cap B^c$ using $U = \{1, \dots, 8\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$.

A6. (Economic Application) A consumer has income $w = 100$ and faces prices $p_1 = 2$, $p_2 = 5$. Define:

$$A = \{(x_1, x_2) \in \mathbb{R}_+^2 : 2x_1 + 5x_2 \leq 100\}$$

$$B = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1 \geq 10\}$$

Describe $A \cap B$ geometrically. What does this set represent economically?

A7. (Proof) Prove the distributive law: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ using the element-chasing method.

A8. (Supremum and Infimum) For each set, find $\sup$, $\inf$, and state whether $\max$ and $\min$ exist:

- (a) $A = \{\frac{n}{n+1} : n \in \mathbb{N}\}$
- (b) $B = (-3, 5]$
- (c) $C = \{x \in \mathbb{R} : x^2 < 9\}$
- (d) $D = \{x \in \mathbb{R} : x^3 \leq 8\}$

7.2 Section B: Functions

B1. (Domain and Range) Find the natural domain and range of:

(a) $f(x) = \sqrt{9 - x^2}$

(b) $g(x) = \frac{x - 1}{x^2 - 4}$

(c) $h(x) = \ln(x^2 - 1)$

(d) $k(x, y) = \sqrt{x} + \sqrt{y}$ (treat as a function from $\mathbb{R}^2$)

B2. (Injectivity and Surjectivity) For each function, determine whether it is injective, surjective, bijective, or none:

(a) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 3x - 7$

(b) $g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = x^3 - x$

(c) $h : \mathbb{R} \rightarrow \mathbb{R}^+, h(x) = e^x$

(d) $k : [0, \pi] \rightarrow [-1, 1], k(x) = \cos x$

(e) $\ell : \mathbb{R}_+ \rightarrow \mathbb{R}_+, \ell(x) = x^3$

B3. (Inverse Functions) Find the inverse of each function and state its domain:

(a) $f(x) = 5x + 3$

(b) $g(x) = \frac{x + 1}{x - 2}, x \neq 2$

(c) $h(x) = e^{2x-1}$

(d) $k(x) = x^3 + 2$ on $\mathbb{R}$

B4. (Composition) Let $f(x) = 2x^2 - 1$ and $g(x) = \sqrt{x + 3}$ (defined on $[-3, \infty)$). Compute:

(a) $(g \circ f)(x)$ and state its domain.

(b) $(f \circ g)(x)$ and state its domain.

(c) $(g \circ f)(2)$ and $(f \circ g)(1)$.

(d) Are $g \circ f$ and $f \circ g$ equal? What does this illustrate?

B5. (Pre-image) Let $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2 - 4$. Find:

(a) $f^{-1}(\{0\})$ (b) $f^{-1}([-4, 0])$ (c) $f^{-1}((0, 5])$ (d) $f^{-1}(\{-5\})$

B6. (Homogeneity) Determine whether each function is homogeneous, and if so, find the degree:

(a) $f(K, L) = 3K^{0.4}L^{0.6}$

(b) $g(x, y) = x^2 + xy + y^2$

(c) $h(x, y) = \frac{x^2y}{x^3 + y^3}$

(d) $k(x, y) = x + y + 1$

B7. (Concavity and Convexity) Determine whether each function is convex, concave, or neither:

(a) $f(x) = x^4$ on $\mathbb{R}$

(b) $g(x) = x^3$ on $\mathbb{R}$

(c) $h(x) = -2x^2 + 4x - 1$ on $\mathbb{R}$

(d) $k(x) = |x|$ on $\mathbb{R}$

B8. (Economic Application — Full Question) A monopolist faces inverse demand $P(Q) = 100 - 2Q$ and has total cost $C(Q) = Q^2 + 10Q + 50$.

(a) Define the profit function $\pi(Q)$ and state its domain.

(b) Find the range of π .

(c) Is π concave? Justify.

(d) Find the profit-maximising quantity and maximum profit.

(e) Find $\pi^{-1}(\{0\})$ (the break-even quantities) and interpret.

8 Full Solutions

8.1 Solutions: Section A

Solution

A1. Set Builder Notation

(a) Even natural numbers less than 20:

$$A = \{x \in \mathbb{N} : x \text{ is even and } x < 20\} = \{2, 4, 6, 8, 10, 12, 14, 16, 18\}. \quad \text{Finite (9 elements).}$$

(b) Real numbers with $x^2 < 16$:

$$B = \{x \in \mathbb{R} : x^2 < 16\} = \{x \in \mathbb{R} : |x| < 4\} = (-4, 4). \quad \text{Infinite.}$$

(c) Prices with $D(P) = 50 - P > 0$ and $P \in [0, 100]$:

$D(P) > 0 \Rightarrow P < 50$. Combined with $P \in [0, 100]$:

$$C = \{P \in [0, 100] : 50 - P > 0\} = [0, 50). \quad \text{Infinite.}$$

Solution

A2. Set Operations

With $U = \{1, \dots, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 5, 6, 7\}$, $C = \{5, 6, 7, 8, 9\}$:

(a) $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$

(b) $A \cap B = \{3, 4, 5\}$

(c) $A \setminus B = \{1, 2\}$ (elements in A not in B)

(d) $B \setminus A = \{6, 7\}$ (elements in B not in A)

(e) $(A \cup B)^c = U \setminus \{1, 2, 3, 4, 5, 6, 7\} = \{8, 9, 10\}$

(f) $A \cap B \cap C = \{3, 4, 5\} \cap \{5, 6, 7, 8, 9\} = \{5\}$

(g) $B \cap C = \{5, 6, 7\}$, so $A \cup (B \cap C) = \{1, 2, 3, 4, 5\} \cup \{5, 6, 7\} = \{1, 2, 3, 4, 5, 6, 7\}$

Check distributivity: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, $A \cup C = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. $(A \cup B) \cap (A \cup C) = \{1, 2, 3, 4, 5, 6, 7\}$. ✓

Solution

A3. Power Set

$A = \{x, y\}$, so $|A| = 2$ and $|\mathcal{P}(A)| = 2^2 = 4$:

$$\mathcal{P}(A) = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}$$

Now $|\mathcal{P}(A)| = 4$, so $|\mathcal{P}(\mathcal{P}(A))| = 2^4 = 16$.

Solution

A4. Cartesian Product

$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$ ($|A \times B| = 2 \times 3 = 6$ elements)

$B \times A = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}$ ($|B \times A| = 3 \times 2 = 6$ elements)

$A \times B \neq B \times A$: the pairs are ordered differently; e.g. $(1, a) \in A \times B$ but $(1, a) \notin B \times A$ since $1 \notin B$. In general $A \times B = B \times A$ only if $A = B$.

Solution

A5. De Morgan's Law Verification

$U = \{1, \dots, 8\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$.

Left side: $A \cup B = \{1, 2, 3, 4, 5, 6\}$, so $(A \cup B)^c = \{7, 8\}$.

Right side: $A^c = \{5, 6, 7, 8\}$, $B^c = \{1, 2, 7, 8\}$, so $A^c \cap B^c = \{7, 8\}$.
 $(A \cup B)^c = A^c \cap B^c = \{7, 8\}$. ✓

Solution

A6. Budget Set Intersection

$A = \{(x_1, x_2) \in \mathbb{R}_+^2 : 2x_1 + 5x_2 \leq 100\}$ is the full budget set (triangle with vertices $(0, 0)$, $(50, 0)$, $(0, 20)$).

$B = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1 \geq 10\}$ is the closed half-plane to the right of $x_1 = 10$.

$A \cap B = \{(x_1, x_2) \in \mathbb{R}_+^2 : 2x_1 + 5x_2 \leq 100, x_1 \geq 10\}$

Geometrically: the budget triangle *truncated* by the vertical line $x_1 = 10$. It is a quadrilateral with vertices $(10, 0)$, $(50, 0)$, $(0, 20)$... wait — at $x_1 = 10$: $2(10) + 5x_2 \leq 100 \Rightarrow x_2 \leq 16$. So the binding vertices are $(10, 0)$, $(50, 0)$, and $(10, 16)$.

Economic interpretation: affordable bundles that contain *at least 10 units of good 1* (perhaps a minimum consumption requirement or a rationing constraint). This is the feasible set when the consumer is forced to consume at least 10 units of good 1.

Solution

A7. Proof of Distributive Law

We prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ by element-chasing.

$(\subseteq)$: Let $x \in A \cap (B \cup C)$. Then $x \in A$ and $x \in B \cup C$.

Case 1: $x \in B$. Then $x \in A$ and $x \in B$, so $x \in A \cap B \subseteq (A \cap B) \cup (A \cap C)$.

Case 2: $x \in C$. Then $x \in A$ and $x \in C$, so $x \in A \cap C \subseteq (A \cap B) \cup (A \cap C)$.

In either case, $x \in (A \cap B) \cup (A \cap C)$.

$(\supseteq)$: Let $x \in (A \cap B) \cup (A \cap C)$.

Case 1: $x \in A \cap B$. Then $x \in A$ and $x \in B \subseteq B \cup C$, so $x \in A \cap (B \cup C)$.

Case 2: $x \in A \cap C$. Then $x \in A$ and $x \in C \subseteq B \cup C$, so $x \in A \cap (B \cup C)$.

Since $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ and $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$, equality holds. $\square$

Solution

A8. Supremum and Infimum

(a) $A = \left\{ \frac{n}{n+1} : n \in \mathbb{N} \right\} = \left\{ \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots \right\}$

The sequence $n/(n+1)$ is strictly increasing and approaches 1. So $\sup A = 1$ (not attained: $n/(n+1) < 1$ for all n , so $\max A$ does not exist). $\inf A = \min A = \frac{1}{2}$ (attained at $n = 1$).

(b) $B = (-3, 5]$: $\inf B = -3$ (not attained, $\min B$ DNE); $\sup B = \max B = 5$.

- (c) $C = \{x \in \mathbb{R} : x^2 < 9\} = (-3, 3)$: $\inf C = -3$ (DNE); $\sup C = 3$ (DNE). Neither min nor max exists.
- (d) $D = \{x \in \mathbb{R} : x^3 \leq 8\} = (-\infty, 2]$: $\sup D = \max D = 2$. D is unbounded below: $\inf D = -\infty$ (no infimum in $\mathbb{R}$).

8.2 Solutions: Section B

Solution

B1. Domain and Range

(a) $f(x) = \sqrt{9 - x^2}$: need $9 - x^2 \geq 0$, i.e. $x^2 \leq 9$, i.e. $x \in [-3, 3]$.

Range: $f(x) \geq 0$ and $f(0) = 3$, so range = $[0, 3]$.

(b) $g(x) = \frac{x-1}{x^2-4} = \frac{x-1}{(x-2)(x+2)}$: need $x \neq \pm 2$.

Domain = $\mathbb{R} \setminus \{-2, 2\}$. Range: g takes all real values except... at $x = 1$, $g(1) = 0$. Analysis shows range = $\mathbb{R}$ (every value achieved by continuity and the behaviour near the poles).

(c) $h(x) = \ln(x^2 - 1)$: need $x^2 - 1 > 0$, i.e. $|x| > 1$.

Domain = $(-\infty, -1) \cup (1, \infty)$. As $|x| \rightarrow 1^+$, $\ln(x^2 - 1) \rightarrow -\infty$; as $|x| \rightarrow \infty$, $\ln(x^2 - 1) \rightarrow \infty$. Range = $\mathbb{R} = (-\infty, \infty)$.

(d) $k(x, y) = \sqrt{x} + \sqrt{y}$: need $x \geq 0$ and $y \geq 0$.

Domain = $\mathbb{R}_+^2 = [0, \infty)^2$. Range: $k(x, y) \geq 0$ always; for any $z \geq 0$, take $x = z^2$, $y = 0$: $k(z^2, 0) = z$. Range = $[0, \infty)$.

Solution

B2. Injectivity and Surjectivity

(a) $f(x) = 3x - 7$: **Bijjective.**

Injective: $3x_1 - 7 = 3x_2 - 7 \Rightarrow x_1 = x_2$. ✓

Surjective: for any $y \in \mathbb{R}$, $x = (y + 7)/3$ gives $f(x) = y$. ✓

(b) $g(x) = x^3 - x = x(x - 1)(x + 1)$: **Neither.**

Not injective: $g(0) = 0 = g(1)$... let's check: $g(0) = 0$, $g(-1) = (-1)(0) = 0$ wait: $g(-1) = (-1)(-2)(0) = 0$. So $g(0) = g(-1) = 0$ with $0 \neq -1$. Not injective.

Not surjective? Actually it IS surjective since $\lim_{x \rightarrow \pm\infty} g(x) = \pm\infty$ and g is continuous, so by IVT every value in $\mathbb{R}$ is achieved. So g is **surjective but not injective**.

(c) $h : \mathbb{R} \rightarrow \mathbb{R}^+$, $h(x) = e^x$: **Bijjective.**

Injective: $e^{x_1} = e^{x_2} \Rightarrow x_1 = x_2$. ✓

Surjective (onto $\mathbb{R}^+$): for any $y > 0$, $x = \ln y \in \mathbb{R}$ gives $h(x) = e^{\ln y} = y$. ✓

(d) $k : [0, \pi] \rightarrow [-1, 1]$, $k(x) = \cos x$: **Bijjective.**

Injective: $\cos$ is strictly decreasing on $[0, \pi]$, so injective. ✓

Surjective (onto $[-1, 1]$): $k(0) = 1$, $k(\pi) = -1$, continuous, so by IVT every value in $[-1, 1]$ is achieved. ✓

(e) $\ell : \mathbb{R}_+ \rightarrow \mathbb{R}_+$, $\ell(x) = x^3$: **Bijjective.**

Injective: x^3 is strictly increasing on $\mathbb{R}_+$. ✓

Surjective: for any $y \geq 0$, $x = y^{1/3} \geq 0$ gives $\ell(x) = y$. ✓

Solution**B3. Inverse Functions**

(a) $f(x) = 5x + 3$: Set $y = 5x + 3 \Rightarrow x = \frac{y-3}{5}$.

So $f^{-1}(y) = \frac{y-3}{5}$, domain $\mathbb{R}$.

(b) $g(x) = \frac{x+1}{x-2}$, $x \neq 2$: Set $y = \frac{x+1}{x-2}$.

$y(x-2) = x+1 \Rightarrow xy - 2y = x+1 \Rightarrow x(y-1) = 2y+1 \Rightarrow x = \frac{2y+1}{y-1}$.

So $g^{-1}(y) = \frac{2y+1}{y-1}$, domain $\mathbb{R} \setminus \{1\}$.

(c) $h(x) = e^{2x-1}$: Set $y = e^{2x-1} \Rightarrow \ln y = 2x-1 \Rightarrow x = \frac{\ln y + 1}{2}$.

So $h^{-1}(y) = \frac{\ln y + 1}{2}$, domain $(0, \infty)$.

(d) $k(x) = x^3 + 2$ on $\mathbb{R}$: Set $y = x^3 + 2 \Rightarrow x^3 = y - 2 \Rightarrow x = (y - 2)^{1/3}$.
So $k^{-1}(y) = (y - 2)^{1/3}$, domain $\mathbb{R}$ (cube roots defined for all reals).

Solution**B4. Composition**

$f(x) = 2x^2 - 1$, $g(x) = \sqrt{x+3}$ with domain $[-3, \infty)$.

(a) $(g \circ f)(x) = g(f(x)) = \sqrt{(2x^2 - 1) + 3} = \sqrt{2x^2 + 2} = \sqrt{2} \cdot \sqrt{x^2 + 1}$.

Domain: need $f(x) \in [-3, \infty)$, i.e. $2x^2 - 1 \geq -3$, i.e. $x^2 \geq -1$. This holds for all $x \in \mathbb{R}$.

Domain of $g \circ f$ is $\mathbb{R}$.

(b) $(f \circ g)(x) = f(g(x)) = 2(\sqrt{x+3})^2 - 1 = 2(x+3) - 1 = 2x + 5$.

Domain: need $x \in [-3, \infty)$ (the domain of g).

Domain of $f \circ g$ is $[-3, \infty)$.

(c) $(g \circ f)(2) = \sqrt{2(4) + 2} = \sqrt{10}$

$(f \circ g)(1) = 2(1) + 5 = 7$

(d) $(g \circ f)(x) = \sqrt{2}\sqrt{x^2 + 1}$ vs. $(f \circ g)(x) = 2x + 5$. These are clearly different functions.

This illustrates that function composition is **not commutative**: $g \circ f \neq f \circ g$ in general.

Solution**B5. Pre-image**

$f(x) = x^2 - 4$.

(a) $f^{-1}(\{0\}) = \{x : x^2 - 4 = 0\} = \{x : x^2 = 4\} = \{-2, 2\}$.

(b) $f^{-1}([-4, 0]) = \{x : -4 \leq x^2 - 4 \leq 0\} = \{x : 0 \leq x^2 \leq 4\} = \{x : |x| \leq 2\} = [-2, 2]$.

(c) $f^{-1}((0, 5]) = \{x : 0 < x^2 - 4 \leq 5\} = \{x : 4 < x^2 \leq 9\} = \{x : 2 < |x| \leq 3\} = [-3, -2) \cup (2, 3]$.

(d) $f^{-1}(\{-5\}) = \{x : x^2 - 4 = -5\} = \{x : x^2 = -1\} = \emptyset$.

(No real number has a negative square, so no input maps to -5 .)

Solution**B6. Homogeneity**

(a) $f(K, L) = 3K^{0.4}L^{0.6}$:

$$f(tK, tL) = 3(tK)^{0.4}(tL)^{0.6} = 3t^{0.4}K^{0.4}t^{0.6}L^{0.6} = t^1 \cdot 3K^{0.4}L^{0.6} = t^1 f(K, L).$$

Homogeneous of degree 1 (CRS). By Euler's theorem: $K \cdot \frac{\partial f}{\partial K} + L \cdot \frac{\partial f}{\partial L} = 1 \cdot f(K, L)$.

(b) $g(x, y) = x^2 + xy + y^2$:

$$g(tx, ty) = (tx)^2 + (tx)(ty) + (ty)^2 = t^2x^2 + t^2xy + t^2y^2 = t^2(x^2 + xy + y^2) = t^2g(x, y).$$

Homogeneous of degree 2.

(c) $h(x, y) = \frac{x^2y}{x^3 + y^3}$:

$$h(tx, ty) = \frac{(tx)^2(ty)}{(tx)^3 + (ty)^3} = \frac{t^3x^2y}{t^3(x^3 + y^3)} = \frac{x^2y}{x^3 + y^3} = t^0 \cdot h(x, y).$$

Homogeneous of degree 0 (like Walrasian demand: scale-invariant).

(d) $k(x, y) = x + y + 1$:

$$k(tx, ty) = tx + ty + 1 = t(x + y) + 1 \neq t^r(x + y + 1) \text{ for any } r \text{ (the constant 1 is unaffected by scaling).}$$

Not homogeneous of any degree.

Solution**B7. Convexity and Concavity**

(a) $f(x) = x^4$: $f''(x) = 12x^2 \geq 0$ for all x , with equality only at $x = 0$.

Convex (not strictly convex everywhere, since $f''(0) = 0$, but it is strictly convex in the sense that $f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y)$ for $x \neq y$).

(b) $g(x) = x^3$: $g''(x) = 6x$. $g''(x) > 0$ for $x > 0$ (convex) and $g''(x) < 0$ for $x < 0$ (concave). **Neither convex nor concave** on all of $\mathbb{R}$.

(c) $h(x) = -2x^2 + 4x - 1$: $h''(x) = -4 < 0$ for all x . **Strictly concave.**

(d) $k(x) = |x|$: For $x \geq 0$, $k'' = 0$ (weakly convex); for $x \leq 0$, $k'' = 0$. Using the definition: $k(\lambda x + (1 - \lambda)y) \leq \lambda k(x) + (1 - \lambda)k(y)$ by the triangle inequality. $|\lambda x + (1 - \lambda)y| \leq |\lambda x| + |(1 - \lambda)y| = \lambda|x| + (1 - \lambda)|y|$ (since $\lambda, 1 - \lambda \geq 0$). **Convex** (weakly, not strictly).

Solution**B8. Monopolist Economic Application**

(a) Profit function:

$$\pi(Q) = \text{Revenue} - \text{Cost} = P(Q) \cdot Q - C(Q) = (100 - 2Q)Q - (Q^2 + 10Q + 50)$$

$$\pi(Q) = 100Q - 2Q^2 - Q^2 - 10Q - 50 = -3Q^2 + 90Q - 50$$

Domain: $Q \geq 0$ and $P(Q) = 100 - 2Q \geq 0 \Rightarrow Q \leq 50$. So domain = $[0, 50]$.

(b) $\pi(Q) = -3Q^2 + 90Q - 50$ is a downward parabola. Maximum at $Q^* = 15$ (see (d)):

$$\pi_{\max} = -3(225) + 90(15) - 50 = -675 + 1350 - 50 = 625$$

At endpoints: $\pi(0) = -50$, $\pi(50) = -3(2500) + 4500 - 50 = -50$.

Range = $[-50, 625]$.

(c) $\pi''(Q) = -6 < 0$ for all Q . Therefore π is **strictly concave**.

Economic meaning: each additional unit of output raises profit by less and less (eventually reducing it). The profit-maximising Q is uniquely determined and is a global maximum.

(d) First-order condition: $\pi'(Q) = -6Q + 90 = 0 \Rightarrow Q^* = 15$.

$\pi(15) = -3(225) + 90(15) - 50 = -675 + 1350 - 50 = \mathbf{625}$.

Maximum profit is **\$625** at output $Q^* = 15$.

(e) Break-even: $\pi(Q) = 0$:

$$-3Q^2 + 90Q - 50 = 0 \Rightarrow 3Q^2 - 90Q + 50 = 0$$

$$Q = \frac{90 \pm \sqrt{8100 - 600}}{6} = \frac{90 \pm \sqrt{7500}}{6} = \frac{90 \pm 50\sqrt{3}}{6} = 15 \pm \frac{25\sqrt{3}}{3}$$

$$Q_1 = 15 - \frac{25\sqrt{3}}{3} \approx 15 - 14.43 = 0.57, \quad Q_2 = 15 + \frac{25\sqrt{3}}{3} \approx 29.43$$

So $\pi^{-1}(\{0\}) \approx \{0.57, 29.43\}$.

Interpretation: The firm breaks even at approximately $Q = 0.57$ (barely above shutdown) and $Q = 29.43$ (the high-output break-even point). For $Q \in (0.57, 29.43)$, the firm makes positive profit; outside this range, it makes a loss. The firm should produce $Q^* = 15$ to maximise profit.

Summary: Key Definitions and Results

Concept	Statement / Formula
Set equality	$A = B \iff (x \in A \iff x \in B)$
Subset	$A \subseteq B \iff (x \in A \Rightarrow x \in B)$
Power set size	$ A = n \Rightarrow \mathcal{P}(A) = 2^n$
De Morgan	$(A \cup B)^c = A^c \cap B^c; \quad (A \cap B)^c = A^c \cup B^c$
Function	$f : X \rightarrow Y$ assigns exactly one $f(x) \in Y$ to each $x \in X$
Injective	$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$
Surjective	$\forall y \in Y, \exists x \in X : f(x) = y$
Bijjective	Injective + surjective; invertible
Inverse	Exists $\iff f$ bijective; $f^{-1} \circ f = \text{id}_X$
Composition	$(g \circ f)(x) = g(f(x));$ not commutative in general
Pre-image	$f^{-1}(B) = \{x \in X : f(x) \in B\}$ (defined for all f)
Concave	$f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$
Convex	$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$
Homogeneous deg. k	$f(tx) = t^k f(x)$ for all $t > 0$
Euler's theorem	$\sum_i x_i \frac{\partial f}{\partial x_i} = k \cdot f(x)$
Fixed point	$f(x^*) = x^*$; guaranteed by Brouwer if $f : C \rightarrow C$ continuous, C compact convex
Supremum / Infimum	Least upper bound / greatest lower bound; may not be attained

End of Notes — ECON 320 Sets and Functions—Fall 2026